

Dialect-Aware, Zero-Token Phonics RAG

Acoustic Accent Normalization, Phonemic Vector Maps & pgvector Retrieval

EdTech Architecture Systems Design Group

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Executive Architecture Synthesis: Early literacy engines face a two-fold engineering dilemma: unadapted acoustic aligners misdiagnose regional accents as phonemic reading deficits, while generative LLMs are too slow ($> 1.5s$), costly, and prone to hallucinating out-of-phase vocabulary. This unified specification demonstrates how **Acoustic G2P Normalization** decouples dialect from decoding error, while **pgvector (Supabase)** serves as the semantic bridge for peer clustering and targeted decodable retrieval—executing in **$<15ms$ at 0 runtime LLM tokens**.

1 The Core Challenge: The Acoustic and Dialect Trap

A Vector DB (pgvector in Supabase) does not replace raw forced alignment; tools like Montreal Forced Aligner (MFA), Kaldi, or Wav2Vec2/CTC operate directly on raw audio spectrograms. However, naive acoustic alignment paired with standard Received Pronunciation (RP) or General American (GA) baselines introduces severe systemic biases:

- **The TRAP-BATH Split:** In RP, "bath" is pronounced /b-ah-th/. In Northern England, Scotland, and Northern Ireland, it is produced with a short vowel as /b-ae-th/. Standard models penalize these children for missing long vowel digraphs.
- **Rhoticity Variance:** Ulster, Scots, and GA dialects fully articulate post-vocalic /r/ ("park" → /p-aa-r-k/). Non-rhotic acoustic graphs flag the rhotic consonant as an intrusive speech anomaly.
- **Diphthong Mergers and Fronting:** Regional vocalic realizations (e.g., centralized diphthongs in Belfast or the *cot-caught* merger) register as phonemic substitutions in unadapted Grapheme-to-Phoneme (G2P) dictionaries.

Dialect normalization must occur **upstream** of the vector layer. This guarantees that the dense learner vector reflects genuine decoding competence rather than regional phonetics.

End-to-End Accent-Aware Processing Flow

[Raw Audio: "The knight parked in the grass"]



Layer 1: Multi-Pronunciation Acoustic Aligner (CTC / MFA)

Evaluates speech energy against regional G2P allophone OR-graphs



Layer 2: Phonological Normalization Filter

Separates acceptable dialect shifts from untaught decoding errors



Layer 3: Dense Mastery Vector Update ($\vec{v}_{\text{student}}$)

Maintains pristine phonics competency vector; clusters cohort peers in pgvector



Layer 4: Dialect-Constrained Decodable RAG (Supabase)

Retrieves pre-vetted decodable texts matching phonics phase and regional safety

2 Layers 1 & 2: Upstream Acoustic Normalization

2.1 Multi-Pronunciation Lexicons (OR-Graphs)

Rather than evaluating acoustic energy against a single phonetic standard, the lexicon defines an OR-graph of legitimate dialectal variants. If the child's acoustic realization traverses any legitimate regional path, the word is scored as acoustically correct:

Listing 1: Regional G2P Dictionary Entries (MFA / Kaldi Lexicon Format)

```
; Lexicon entries accommodating UK, Irish, and Scottish regional allophones
grass  G R AE S      ; Northern English / Ulster / Scottish short-a
grass  G R AA S      ; RP / Southern British broad-a
park   P AA K        ; Non-rhotic standard (RP)
park   P AA R K      ; Rhotic realization (Ulster / Scottish / US)
cow    K AW          ; Standard diphthong
cow    K EU          ; Fronted diphthong realization (Belfast / Antrim)
```

2.2 The Deterministic Normalization Filter

Before updating the student profile vector, an upstream deterministic rule filter isolates dialectal phonological variations from true decoding deficits:

$$\text{Normalized Mastery}(g) = \begin{cases} 1.0 & \text{if realization matches an acceptable regional allophone} \\ 0.0 & \text{if error constitutes an illegal grapheme/phoneme substitution} \end{cases} \quad (1)$$

If a child in Belfast reads *"The knight parked in the grass"*, pronouncing *"grass"* with /ae/ and *"parked"* with rhotic /ɹ/, both are classified as valid regional variants (1.0). If they drop the silent *"k"* in *"knight"* or confuse the *"igh"* trigraph, the normalization filter flags this as an untaught graphemic gap (0.0).

3 Layer 3: Dense Competency Vectors & Cohort Search

Children do not fail reading through isolated, random errors; they struggle with specific structural categories (e.g., stopping fricatives, dropping silent graphemes, or confusing split digraphs).

3.1 High-Dimensional Competency Embeddings

Student phonics mastery is modeled as a normalized dense vector $\vec{v}_{\text{student}} \in \mathbb{R}^D$, where each dimension corresponds to a curriculum phonics unit:

$$\vec{v}_{\text{student}} = [0.95_{\text{CVC}}, 0.90_{\text{short_a}}, 0.88_{\text{blends}}, 0.42_{\text{silent_k}}, 0.35_{\text{igh_trigraph}}, \dots] \quad (2)$$

Because dialect variation is neutralized at Layer 2, this vector reflects purely pedagogical mechanics.

3.2 Automated Peer Cohort Clustering in pgvector

Using cosine distance, educators and intervention engines cluster students sharing identical phonological profiles without manual auditing:

Listing 2: Supabase pgvector: Finding Peer Cohorts with Identical Phonetic Gaps

```
-- Find peers sharing a student's phonetic struggle vector
SELECT
```

```
    student_id,  
    student_name,  
    1 - (phonemic_mastery_vector <=> current_student_vector) AS similarity_score  
FROM student_phonics_profiles  
WHERE class_id = 'primary-3-belfast'  
ORDER BY phonemic_mastery_vector <=> current_student_vector ASC  
LIMIT 4;
```

4 Layer 4: Dialect-Constrained Decodable RAG

The primary operational failure of generative LLMs in early phonics is unpredictability: they introduce irregular vocabulary outside the child's learned phase and suffer from > 1.5s latency.

Our architecture pre-indexes a vast catalog of human-curated decodable sentences. Each chunk is tagged with both phonics phase constraints and regional dialect safety metadata.

4.1 Decodable Passage Schema with Dialect Safety

Passages that rely on specific homophones or rhymes (e.g., rhyming *"bath"* with *"laugh"*, or pairing *"cot"* and *"caught"*) are restricted to compatible dialects:

Listing 3: Decodable Chunk Metadata Structure

```
{
  "passage_id": "knight_quest_01",
  "text": "The brave knight rode high in the night.",
  "target_graphemes": ["kn", "igh"],
  "book_band": "Green",
  "phonics_phase": 5,
  "dialect_compatibility": ["UK_GENERAL", "NI_ULSTER", "SCOT", "US_GENERAL"]
}
```

4.2 Sub-15ms Dynamic Retrieval Query

When a student demonstrates a gap on target graphemes (e.g., kn and igh), Supabase retrieves pre-vetted decodables maximizing practice for that specific gap while enforcing strict phase and dialect constraints:

Listing 4: Dialect-Filtered Decodable Retrieval via pgvector

```
-- Retrieve decodables matching child gap vector while enforcing dialect safety
SELECT passage_id, text, book_band
FROM decodable_passages
WHERE phonics_phase <= 5
      AND 'NI_ULSTER' = ANY(dialect_compatibility)
ORDER BY passage_embedding <=> current_student_vector
LIMIT 3;
```

Latency Guarantee: This index query returns candidate decodables in <15ms, delivering culturally safe, pedagogical targeted practice at **0 LLM token cost**.

5 Unified System Architecture & Cost Model

6 Key Architectural Takeaways

1. **Acoustics Absorb Accents, Vectors Store Mastery:** Neutralizing accent variations upstream ensures the competency vector represents genuine phonics decoding skill, eliminating diagnostic bias.
2. **Zero Generative Hallucinations:** Pre-vetting decodable chunks guarantees out-of-phase phonemes and illegal irregular words never reach early readers.
3. **Sub-15ms, 0-Token Performance:** 99% of real-time student reading interactions are powered entirely by deterministic filtering and vector indexing, keeping operating costs near zero.

Layer	Component	Core Functionality	Runtime Cost
1. Acoustic	Speechmatics / MFA / CTC	Aligns speech energy using regional G2P multi-pronunciation lexicons	0 Tokens
2. Normalization	Deterministic Filter	Separates dialectal shifts from true graphemic decoding errors	0 Tokens
3. Vector Store	Supabase (pgvector)	Maintains $\vec{v}_{\text{student}}$ and clusters cohort intervention groups	0 Tokens
4. Decodable RAG	Curated Passage DB	Matches decodables constrained by phonics phase and dialect tags	0 Tokens
5. Async Diagnostic	LLM Agent (Lang-Graph)	Batches weekly phonetic progress into actionable teacher summaries	Sub-cent

Table 1: End-to-End Layer Boundaries, Component Responsibilities, and Token Economics